SUMMARY

This project implements a deep learning system for classifying Alzheimer's disease severity from MRI scans. Using transfer learning with ResNet50, the model achieved 97.8% accuracy on test data, successfully distinguishing between four clinical stages: NonDemented, VeryMildDemented, MildDemented, and ModerateDemented. The project also uncovered important quantitative features showing progressive intensity changes in brain scans as Alzheimer's advances.

PROJECT OBJECTIVES

- Develop an accurate deep learning model for Alzheimer's stage classification
- Extract and analyze quantitative features from MRI images
- Provide a reproducible pipeline for medical image analysis
- Demonstrate practical application of transfer learning in healthcare

DATASET DESCRIPTION

Source: Kaggle Alzheimer's Dataset (4-class MRI collection)

Total Images: 44,000 MRI scans

Class Distribution:

- NonDemented: 19,200 images (43.6%)
- VeryMildDemented: 16,800 images (38.2%)
- MildDemented: 15,000 images (34.1%)
- ModerateDemented: 15,000 images (34.1%)

DATA PROCESSING PIPELINE

- Corrupted image detection: Scanned 200 images per class, found 0 corrupted
- Dimension consistency: Majority images at 176×208 pixels
- Class balance: Used stratified sampling to maintain distribution

PREPROCESSING

Image resizing to 192×192 pixels

Color space conversion (BGR to RGB)

Data augmentation for training set

ResNet50-specific preprocessing

Stratified split: 70% train, 15% validation, 15% test

FEATURE ENGINEERING AND ANALYSIS

We extracted numerical features from 50 random images per class using this function:

Key Feature Findings

Intensity Analysis Results

Class	Mean Intensity	Std Intensity	Gray Mean	Gray Std
NonDemented	77.5	82.4	77.5	82.1
VeryMildDemented	75.9	81.2	75.9	81.2
MildDemented	70.6	80.7	70.2	80.8
ModerateDemented	69.5	78.0	68.9	78.1

Observations

Progressive Intensity Decrease:

Healthy brains: 77.5 brightness

- Advanced Alzheimer's: 69.5 brightness
- 10.3% reduction from healthy to severe cases

Early Detection Signal:

- VeryMildDemented shows 2.1% intensity drop
- Provides measurable early warning sign

Texture Changes:

- Standard deviation decreases with disease severity
- Suggests tissue becomes more homogeneous in advanced stages

MODEL ARCHITECTURE

Transfer Learning Implementation

We used ResNet50 pre-trained on ImageNet with custom classification layers

MODEL EVALUATION

Detailed Classification Report

Class	Precision	Recall	F1-Score	Support
MildDemented	0.969	0.996	0.982	1500
ModerateDemented	0.999	0.999	0.999	1500
NonDemented	0.992	0.944	0.967	1920
VeryMildDemented	0.955	0.984	0.969	1680

Overall Accuracy: 97.8% on 6,600 test image

Test Results:

- Test Accuracy: 97.8% (0.9785)

• Test Loss: 0.0673

PROJECT INSIGHT

• Transfer learning with ResNet50 works exceptionally well for medical images
• Simple numerical features (intensity, texture) provide valuable insights
• Data augmentation significantly improves model generalization
• Early stopping prevents overfitting effective